

Consistency Is Not Coherence: Orientation Search for Certified Alignments Between 4D Defence Upper Ontologies

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Abstract

We align three upper ontologies that sit under UK and NATO defence data infrastructure: the Information Exchange Standard (IES), the Higher Quality Data Model (HQDM) that underpins the National Digital Twin, and Basic Formal Ontology (BFO). No public alignment between IES and HQDM existed. Promoting a hand-curated 17-correspondence crosswalk to OWL and reasoning over the complete merged ontologies with HermiT produces four measurements and one method that we believe matter beyond this pair. First, the published `hqdm.owl` shipped by the UK Government Communications Headquarters (GCHQ) is *natively incoherent*: 39 of its 229 named classes are unsatisfiable before any alignment is added, while IES and BFO have none. An alignment evaluated against it inherits a broken target. Second, naive equivalence promotion leaves the merge *consistent*, so the usual check passes, while creating 100 new unsatisfiable classes and 218 entailed source-internal subsumptions that neither standard asserts. Consistency is the wrong acceptance test for an alignment. Third, we present *orientation search*, a repair operator that treats the direction of each correspondence ($\sqsubseteq$, $\sqsupseteq$, $\equiv$, discard) as the search variable rather than its presence, with a reasoner as oracle for coherence and conservativity. It yields a 21-axiom IES $\leftrightarrow$ HQDM bridge at zero new unsatisfiable classes and zero conservativity violations, identical under all 200 correspondence orders we tried, in which every weakening carries a machine-found counterexample as its justification. Crossing to BFO is possible only through the occurrent branch: with the rest of the bridge fixed, mapping `ies:Entity` to `bfo:material entity` makes 106 IES classes unsatisfiable, weakening that mapping to a subsumption still leaves 106 unsatisfiable, and re-targeting it to `bfo:history` leaves none. The defect is the target, not the strength. Finally, no system we test produces this bridge: LogMap, LogMapLt and two LLM oracles all accept the crosswalk's documented false friend `ies:Event` $\equiv$ `hqdm:event`, and LogMap's repair, given our expert input, is safe but keeps none of the five provable equivalences. All artifacts are open.

Keywords

ontology matching, alignment coherence, alignment repair, conservativity, upper ontologies, 4D ontology, defence data standards

1. Introduction

Alignments are usually published as sets of correspondences with a similarity score, and consumed as if they were logical axioms. The gap between those two readings is where this paper lives, and the setting is deliberately unglamorous. The UK Information Exchange Standard (IES) is the vocabulary in which UK defence and national-security systems exchange assertions about people, events and places. The Higher Quality Data Model (HQDM) [13] is the foundation data model beneath the National Digital Twin and the built-environment estate. Both descend from the same BORO and ISO 15926 4D tradition [14], which is a fact about their lineage and, as this paper shows, no guarantee that their axioms agree. Joining them is nonetheless ordinary operational work: a system reasoning about an activity in a real place has to combine what IES says about the operational picture with what HQDM says about the estate and terrain it happens in. Basic Formal Ontology (BFO) [15, 16] is the third party in the room because it is the realist upper ontology of the US and NATO defence data foundry, so a UK-to-coalition join runs through it. No public machine-readable alignment between IES and HQDM

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existed before this work. For IES and BFO, a thirteen-author team spanning the IES, BORO and BFO communities has recently compared the two frameworks at the level of design patterns [19], explicitly deferring “axiomatic translation definitions” that would move “from plausible semantic equivalence to provable equivalence” to future work. This paper supplies that layer: every correspondence here is a reasoner-certified OWL axiom.

We built one, then asked the question that the ontology matching community has long treated as central and that alignment consumers still routinely skip: what happens when the correspondences are read as logic? The answer, on real published artifacts, is worse than the folklore suggests, and the repair is more interesting than deletion.

Contributions. The paper contributes one method and one publication proposal, and reports four measurements that they produced. We separate the two kinds of claim because a finding about HQDM’s shipped file is not the same sort of thing as a repair operator, and a reader is entitled to know which is which.

1. **Orientation search** (§3), a repair operator whose search variable is the *direction* of each correspondence rather than its presence. Weakening an equivalence to a subsumption is not new and LogMap has done it for over a decade (§8); what is new is searching the lattice ($\equiv$, $\sqsubseteq$, $\sqsupseteq$, discard) together with weakening to an ancestor, with a complete DL reasoner refereeing every step against a coherence and a conservativity criterion and completion back toward equivalence attempted once safety holds. Every rejected direction is returned with the reasoner’s counterexample. §3.3 states what the operator guarantees, which does not include minimality.
2. **Directed, justified alignment publication** (§6). A correspondence published as $\langle s, o, 0.85 \rangle$ discards the difference between “these are the same” and “every s is an o , and here is the individual showing the converse fails”. We argue that the direction should be computed and published with its justification, and that a coherence report on each input ontology belongs with a published matching test case.

The measurements are these. (i) The published `hqdm.owl` is natively incoherent: 39 of its 229 named classes are unsatisfiable before any alignment is added, invisible both in the asserted hierarchy and to a consistency check (§4.1). (ii) Naive equivalence promotion of the expert crosswalk leaves the merge consistent, so it passes the check most pipelines run, while adding 100 new unsatisfiable classes and 218 conservativity violations, the 218 surviving even when HQDM is repaired first (§4.2, §4.1). (iii) Orientation search returns a 21-axiom IES $\leftrightarrow$ HQDM bridge and a four-axiom IES $\leftrightarrow$ BFO bridge at zero new unsatisfiable classes and zero conservativity violations, unchanged across 200 correspondence orders (§4, §4.6), and an ablation (§4.4) separates a mapping that is too strong from one aimed at the wrong target. (iv) LogMap, LogMapLt and two LLM oracles all accept the crosswalk’s documented false friend, and the classes LogMap leaves unsatisfiable prove attributable to an input assumption HQDM violates rather than to incomplete repair (§5).

All ontologies used are public, all code and certified artifacts are released, and the reasoning is deterministic (see the availability statement at the end of the paper).

2. Materials and background

2.1. The three ontologies

Table 1 reports what we measured on the exact vendored files, not what the documentation claims. Two properties are worth noting immediately. IES declares 511 named classes and *zero* class-disjointness axioms; it is a rich taxonomy with almost no exclusion structure. HQDM declares 229 named classes and 14 disjointness axioms, introduced by the EXPRESS-to-OWL rendering. BFO 2020 is small (35 named classes) and heavily constrained (19 disjointness axioms), which is exactly why it is the ontology that punishes a careless mapping hardest.

Table 1

The three ontologies as measured on the vendored files. Named classes counts subjects of `rdf:type owl:Class`, or of `rdf:type rdfs:Class` for IES, which types its classes that way, with an IRI. Disjointness counts `owl:disjointWith` triples; none of the three uses `owl:AllDisjointClasses` or `owl:disjointUnionOf`. Unsatisfiable classes are computed by Hermit on each ontology alone.

Ontology	triples	named classes	object props	disjointness	unsat (alone)
IES (<code>ies-common.ttl</code>)	4,039	511	162	0	0
HQDM (<code>hqdm.owl</code>)	3,127	229	74	14	39
BFO 2020 (<code>bfo.owl</code>)	1,221	35	0	19	0

A note on names: `dst1/IES4` was archived in March 2025 (custodianship moved to the cross-government IES Working Group under DBT, last release v4.3.1) and the canonical successor is `IES-Org/ont-ies`, whose `ies-common.ttl` we vendor. It declares `owl:versionInfo 5.0.3`, so every IES result here is about IES 5.0.3 `ies-common`, not IES4. IES 5 is under active revision and counts differ between lineages, so all IES claims are about the exact file of Table 1, pinned by version IRI and sha256 in Appendix G.

IES and HQDM are both four-dimensionalist: an individual is a spatio-temporal extent, and what a three-dimensionalist calls a state of an object is a temporal part of that extent. BFO is three-dimensionalist at its core, partitioning existence into `bfo:continuant` and `bfo:occurent` and declaring them disjoint. That single disjointness is the whole difficulty of the IES-to-BFO join, and §4.4 shows it is crossable but only in one place.

2.2. The crosswalk

The input alignment is a hand-curated crosswalk of 17 correspondences in SSSOM [9]: 13 `skos:closeMatch` (10 class pairs, 3 property pairs) and 4 `skos:relatedMatch` recorded as warnings rather than mappings, accompanied by six documented divergences, pairs that look like they map and do not. The headline divergence is a false friend any label-based matcher gets exactly backwards: `ies:Event` is a durative happening with participants, `hqdm:event` an instantaneous temporal boundary with no duration and no participants, and the true counterpart of `ies:Event` is `hqdm:activity`. The crosswalk records `ies:Event ~ hqdm:event` at confidence 0.25 purely to carry the warning; every system we test walks straight into it.

2.3. Consistency, coherence, conservativity

We use three standard criteria, and are pedantic about the difference because the central negative result of this paper is that the first is useless here.

An ontology $\mathcal{O}$ is *consistent* if it has a model, and *coherent* if no named class is unsatisfiable ($\mathcal{O} \not\models C \sqsubseteq \perp$ for named C); the distinction is due to Flouris et al. [22]. It is worth stating exactly which direction we rely on. An inconsistent ontology has no model, so every class in it is unsatisfiable; coherence therefore implies consistency for any ontology whose signature contains at least one named class. The qualification matters only in the degenerate case of an ontology declaring no named class at all, which is vacuously coherent and can still be inconsistent, since DL semantics requires a non-empty domain and $\top \sqsubseteq \perp$ then has no model; every merge certified here declares hundreds. The converse fails, and not exotically: a consistent ontology can be satisfied by a model in which a hundred of its classes are simply empty. We apply both criteria to the TBox, and where an alignment is consumed with data the consistency of the merged ABox remains a necessary check that nothing here argues against; our claim is the narrower one that TBox consistency is not a *sufficient* acceptance test for a class-level alignment. Alignment incoherence and its repair are well studied [2, 3, 4, 23], and yet consistency checking remains what deployment pipelines actually run.

The third criterion is *conservativity* [5, 6]. Let $\mathcal{O}_1, \mathcal{O}_2$ be the input ontologies and $\mathcal{M}$ the alignment

read as axioms. $\mathcal{M}$ violates conservativity if the merge entails a subsumption between two classes of the *same* input ontology that the input ontology did not itself entail. Such an entailment is a statement about IES that IES never made, produced by the act of aligning it to something else. A consumer who trusts the merged model is now reasoning with axioms no standards body ever wrote.

Our two acceptance criteria are therefore **S1' (relative coherence)**, that the merge introduces no unsatisfiable class beyond those already unsatisfiable in the inputs alone, and **S2 (conservativity)**, that it entails no new subsumption between two classes of the same input ontology. S1' is relative rather than absolute because HQDM's 39 native casualties (§4.1) would otherwise make every candidate fail identically and carry no signal.

3. Orientation search

3.1. The operator

Alignment repair conventionally computes a diagnosis, a minimal subset of mappings whose removal restores coherence, and deletes it. Deletion throws away a correspondence a domain expert asserted for good reason and records nothing about *why* it failed. Orientation search searches over strength instead: a curated crosswalk gives (s, o) pairs, but because SKOS carries no logical commitment it does not say how strong the relation is, and that is precisely the quantity to search. For each correspondence we consider the lattice

$$\{s \equiv o\} \succ \{s \sqsubseteq o\}, \{s \sqsupseteq o\} \succ \{\emptyset\}$$

and, when even the forward subsumption fails because the target is defective, a fifth move: *weaken to ancestor*, replacing o by a superclass of o , in the spirit of axiom weakening [7].

The search runs in two phases. **P1** accepts each forward edge $s \sqsubseteq o$, weakening the target upward while the edge fails S1' or S2. **P2** greedily attempts each reverse edge $o \sqsubseteq s$; where it survives, the pair completes to a full equivalence. The result declares the standards equivalent exactly where they agree and one-directionally linked exactly where they do not. Algorithm 1 states the procedure, CERTIFY being the oracle that decides S1' and S2.

3.2. Instantiation

The reasoner is HermiT [8] via owlready2, called on the complete merged ontologies, not on fragments and not on hand-selected axiom subsets. The whole script, both bridges and the ablation included, makes on the order of 35 reasoner calls and runs in 33 s wall-clock on the machine in Appendix G. The procedure is deterministic in the sense that matters here: HermiT decides entailment by the hypertableau calculus [8], a decision procedure with no randomised or learned component, so repeated runs on identical inputs return identical certified output. We verified this by re-running the search and diffing the result artifacts, which are byte-identical.

One design decision is worth flagging here: P1 and P2 are both greedy, each correspondence being decided against the axioms already accepted, so the output is in principle a function of the order in which the correspondences arrive. The next subsection states what that does and does not guarantee, and §4.6 measures the order sensitivity rather than arguing about it.

3.3. What the operator guarantees, and what it does not

Orientation search is greedy and contextual, not a diagnosis-based repair, and that distinction fixes what can honestly be claimed for it. The output is sound as a set, since an axiom enters B only after CERTIFY approves all of B with it added and B is certified once more at the end, and termination is immediate, with oracle calls bounded by $\sum_i (\text{depth}(o_i) + 1) + |C| + 2|P|$; the IES $\leftrightarrow$ HQDM run makes 26. There is no minimality guarantee: the operator never computes a diagnosis, so calling its output *maximal* would overstate the case, and §4.6 measures what other orders return. Nor can it reach a correct target off the asserted target's ancestor chain, the BFO case of §4.4. Appendix C gives the guarantees, the

Algorithm 1 Orientation search

Require: input ontologies $\mathcal{O}_1, \mathcal{O}_2$; correspondences C ; reasoner $\mathcal{R}$

```
1:  $U_{\text{nat}} \leftarrow \mathcal{R}.\text{UNSAT}(\mathcal{O}_1 \cup \mathcal{O}_2)$  ▷ native baseline
2:  $B \leftarrow \emptyset$ 
3: for all  $(s, o) \in C$  do ▷ P1: forward, weakening to ancestor
4:    $t \leftarrow o$ 
5:   while  $t \neq \top$  and  $\neg \text{CERTIFY}(B \cup \{s \sqsubseteq t\})$  do
6:      $t \leftarrow \text{PARENT}(t)$ 
7:   end while
8:   if  $t \neq \top$  then  $B \leftarrow B \cup \{s \sqsubseteq t\}$ 
9:   end if
10: end for
11: for all  $(s, o) \in C$  with  $s \sqsubseteq o \in B$  do ▷ P2: completion toward  $\equiv$ 
12:   if  $\text{CERTIFY}(B \cup \{o \sqsubseteq s\})$  then  $B \leftarrow B \cup \{o \sqsubseteq s\}$ 
13:   else record  $\mathcal{R}$ 's witness as the justification for one-directionality
14:   end if
15: end for
16: return  $B$ 

17: function  $\text{CERTIFY}(B)$ 
18:    $U \leftarrow \mathcal{R}.\text{UNSAT}(\mathcal{O}_1 \cup \mathcal{O}_2 \cup B)$ 
19:    $V \leftarrow \{C_1 \sqsubseteq C_2 \text{ entailed by the merge} \mid C_1, C_2 \in \mathcal{O}_i, \mathcal{O}_i \not\models C_1 \sqsubseteq C_2\}$ 
20:   return  $(U \setminus U_{\text{nat}} = \emptyset) \wedge (V = \emptyset)$  ▷ S1' and S2
21: end function
```

comparison with minimal-repair systems [4, 21, 5, 24], the remaining failure modes and the behaviour on coherent inputs.

4. Results

4.1. The target ontology is broken before we touch it

Reasoned alone, `hqdm.owl` has **39 unsatisfiable classes** (Table 1). IES alone and BFO alone have none. The casualties are concentrated in HQDM's relationship-derived branch, including `hqdm:association`, `hqdm:employment`, `hqdm:ownership`, `hqdm:participant`, `hqdm:sign`, `hqdm:asset`, `hqdm:transfer_of_ownership` and `hqdm:physical_quantity`. The mechanism is the collision between the EXPRESS relationship heritage, in which a relationship is a first-class entity, and the `hqdm:class/hqdm:relationship` and `hqdm:abstract_object/hqdm:spatio_temporal_extent` disjointness axioms introduced by the OWL rendering. The file's own header warns of known consistency issues; what we add is the quantification and, more importantly, the identification of *which* classes are affected, because that is what an alignment consumer needs.

Three consequences follow for ontology matching practice. A matcher that maps onto any of those 39 classes has produced a correspondence whose target is provably empty: the mapping is vacuously satisfied and carries no information. A reference alignment built by human curators against this file may contain such mappings, since the defect is invisible in the asserted hierarchy. And an evaluation that reports precision and recall against such a reference is reporting agreement about empty sets. §6 argues from this that a coherence report on each input belongs with a published test case.

Repairing the target first. The natural objection is that HQDM should simply be repaired before anything is aligned to it, and the objection deserves a measurement rather than an argument. We applied

the simplest repair available, deleting all 39 unsatisfiable classes together with every triple mentioning them, which removes 456 of HQDM’s 3,127 triples and 39 of its 229 named classes and does leave the remainder coherent. Three things follow, all measured (`reasoning/hqdm_prerepair.py`). The repair destroys curated content: `hqdm:participant` is one of the 39, so the class `ies:EventParticipant` was mapped to no longer exists and the operator reaches `hqdm:state` only by reading the ancestor chain from the unrepaired file. Naive equivalence promotion of the nine surviving correspondences against the repaired target is still not conservative: new unsatisfiability falls to 0, as it must once the defective class is gone, but the same **218 conservativity violations** remain. And orientation search returns *the identical 21-axiom bridge* on the repaired target that it returns on the published one. The separation this paper reports is therefore not an artifact of HQDM’s incoherence, and neither is the bridge. Pre-repairing the target changes which diagnostic fires, not the verdict, and it costs a correspondence to do it.

4.2. Consistent, and thoroughly incoherent

We promote all 13 `closeMatch` correspondences to `owl:equivalentClass` and `owl:equivalentProperty` and merge the complete ontologies. The merge is **consistent**, so the check deployment pipelines typically run passes, while carrying **100** new unsatisfiable classes beyond HQDM’s native 39 and **218** new source-internal subsumptions.

The 100 casualties are IES operational classes, `ies:Accused`, `ies:Arrested`, `ies:ArrestingOfficer`, `ies:Assessor`, `ies:Witness`, `ies:Customer`, `ies:Authoriser`, `ies:AtWar`: the classes an intelligence or policing application actually instantiates. The 218 conservativity violations include `ies:Entity` $\sqsubseteq$ `ies:State`, `ies:Investigation` $\sqsubseteq$ `ies:State` and `ies:OfferForSale` $\sqsubseteq$ `ies:Entity`, statements about IES in IES’s own vocabulary that IES does not make and that arise solely from having been aligned to HQDM.

Isolation. Exactly one of the ten class targets, `hqdm:participant`, is among HQDM’s 39. Weakening that single edge to `ies:EventParticipant` $\sqsubseteq$ `hqdm:state`, one hop up, takes new unsatisfiability from 100 to 0 (§4.3): one correspondence onto one bad class takes down a hundred classes in the other ontology, and the cost is not local.

4.3. The certified IES $\leftrightarrow$ HQDM bridge

Orientation search returns **21 directed axioms** at 0 new unsatisfiable classes and 0 conservativity violations (Table 2): five class pairs and all three property pairs complete to full equivalence, four class pairs remain one-directional, each carrying the reasoner’s counterexample, and one is weakened.

The witnesses the reasoner returned for those four reverse edges are the scientific content of this result, and Appendix B gives all four in full, each quoted from a sorted violation set so the citation is reproducible rather than an artifact of enumeration order. They are one mechanism seen four times. HQDM asserts `hqdm:activity` $\sqsubseteq$ `hqdm:individual` while IES holds `ies:Event` and `ies:Entity` apart, so reversing `ies:Entity` $\sim$ `hqdm:individual` drags every IES happening under `ies:Entity`: `ies:Arrest` $\sqsubseteq$ `ies:Entity`, and 103 others. HQDM’s `hqdm:individual` is strictly broader than IES’s notion of a whole-life persisting thing, and the other three reversals fail for variants of the same reason.

These are not failures of the crosswalk. They are the two standards disagreeing, localised to a named class by a prover. The `ies:Entity` $\sim$ `hqdm:individual` case says something a human curator can act on: HQDM subsumes activities under individuals, IES insists an event is not an entity, so the two notions of whole-life persisting thing are genuinely different and the correct relation is a subsumption, not an identity. That is a finding about the standards produced as a by-product of certifying the alignment, and it is exactly the kind of content that an alignment published as a bare similarity score cannot carry.

We note the shape of the outcome. Where the two 4D models genuinely agree, the bridge is full equivalence. Where they disagree, it is exactly as weak as the disagreement requires and no weaker. A

Table 2

The certified bridge. Prefixes are omitted: every source is `ies:` and every target `hqdm:`. The last column is the number of S2 violations the *reverse* edge would have introduced, which is why the row is one-directional. Every verdict is a reasoner decision, not a curator’s judgement.

Source (<code>ies:</code>)	Target (<code>hqdm:</code>)	Verdict	S2 if reversed
Element	<code>spatio_temporal_extent</code>	full $\equiv$	0
Thing	<code>thing</code>	full $\equiv$	0
Event	<code>activity</code>	full $\equiv$	0
PossibleWorld	<code>possible_world</code>	full $\equiv$	0
ClassOfElement	<code>class_of_spatio_temporal_extent</code>	full $\equiv$	0
Entity	<code>individual</code>	forward only	104
State	<code>state</code>	forward only	109
PeriodOfTime	<code>period_of_time</code>	forward only	1
ParticularPeriod	<code>period_of_time</code>	forward only	5
EventParticipant	<code>participant $\rightarrow$ state</code>	weakened	n/a
<code>isPartOf</code>	<code>part_of</code>	full $\equiv$	0
<code>isParticipantIn</code>	<code>participant_in</code>	full $\equiv$	0
<code>isStateOf</code>	<code>temporal_part_of</code>	full $\equiv$	0

diagnosis-and-delete repair would have returned a smaller alignment with less information in it.

4.4. Crossing to BFO: only via the occurrent branch

BFO is where the 4D/3D mismatch has to be paid for. Unlike the HQDM case the four BFO targets were specified by hand and then certified: orientation search verified and oriented them but did not discover them, for the reason §3.3 gives. The bridge is

```

ies:Element  $\sqsubseteq$  bfo:occurrent    ies:Entity  $\sqsubseteq$  bfo:history
ies:Event  $\sqsubseteq$  bfo:process        ies:State  $\sqsubseteq$  bfo:occurrent
cwb:hasHistory: bfo:material entity  $\rightarrow$  bfo:history (declared by the bridge)

```

certified at 0 new unsatisfiable classes and 0 conservativity violations. The one axiom in it that a curator would not write is `ies:Entity  $\sqsubseteq$  bfo:history`. The obvious choice is `ies:Entity  $\equiv$  bfo:material entity`, and Table 3 shows what that choice costs by changing *only* that axiom and holding the other three fixed.

Table 3

The cost of one axiom. Every row merges the complete IES and BFO with `ies:Element  $\sqsubseteq$  bfo:occurrent`, `ies:Event  $\sqsubseteq$  bfo:process` and `ies:State  $\sqsubseteq$  bfo:occurrent` held fixed, and varies only how `ies:Entity` is mapped. Counts are IES classes made unsatisfiable.

Mapping for <code>ies:Entity</code>	Rest of bridge	IES classes unsatisfiable
$\equiv$ <code>bfo:material entity</code>	absent	0
$\sqsubseteq$ <code>bfo:material entity</code>	absent	0
$\equiv$ <code>bfo:material entity</code>	present	106
$\sqsubseteq$ <code>bfo:material entity</code>	present	106
$\sqsubseteq$ <code>bfo:history</code>	present	0

Three things follow. First, unsatisfiability here is a property of the bridge as a whole, not of any single mapping. The first two rows of Table 3 are a control rather than a finding: with IES declaring no class disjointness and no other bridge axiom present, a lone correspondence has nothing to contradict. The measurement is the contrast among the last three rows. Only once the rest of the bridge puts IES’s 4D backbone on the occurrent branch does the `bfo:continuant/bfo:occurrent` disjointness fire

and make 106 IES classes unsatisfiable (`ies:Account`, `ies:Actor`, `ies:Aircraft`, `ies:Bank`, ...). An alignment cannot be certified one correspondence at a time.

Second, weakening does not rescue it: a plain subsumption still leaves 106 IES classes unsatisfiable. This is why weaken-to-ancestor is not sufficient on its own, since walking up from `bfo:material entity` reaches `bfo:independent continuant`, `bfo:continuant` and finally `bfo:entity`, where the axiom is safe but vacuous, and the chain never passes through `bfo:history`.

Third, re-targeting rescues it completely. BFO 2020 already ships the class a 4D standard is talking about: `bfo:history` (BFO_0000182), a process that is the totality of what happens to a material entity, sitting natively on the occurrent branch. Changing that one axiom takes unsatisfiable IES classes from 106 to 0. The naive mapping was not too strong, it was aimed at the wrong target, and the distinction matters because only one of those diagnoses is fixed by weakening. A 4D `ies:Entity` is not a BFO continuant; it is the continuant's history, and the bridge declares `cwb:hasHistory` so a consumer can still get from one to the other.

4.5. All three in one artifact

The union of the two bridges over the complete $\text{IES} \cup \text{HQDM} \cup \text{BFO}$ certifies at 0 and 0 and admits five direct $\text{HQDM} \leftrightarrow \text{BFO}$ anchors on top, a fused artifact of 32 subsumption axioms plus `cwb:hasHistory` (Appendix E). That two certified bridges compose without new violations is not guaranteed by their certificates, so composition must in general be re-certified.

4.6. The bridge does not depend on the correspondence order

Because P1 and P2 each decide a correspondence against those already accepted, the returned bridge is in principle a function of the order in which the curator listed the crosswalk, and a greedy procedure with that property deserves to be doubted rather than defended. So we measured it: holding everything else fixed, we re-ran the identical search under 200 random permutations of the ten class correspondences, the first being the curator's confidence-descending order (`reasoning/order_robustness.py`, seed 20260824).

All 200 return the same bridge. There is one distinct output, of 21 directed axioms, minimum and maximum axiom count both 21, at a cost of 970 reasoner calls with a further 4,256 oracle queries served from a memo keyed on the trial axiom set. We do not claim this generalises, and §7 says why a 17-correspondence backbone over an ontology declaring no disjointness is an easy case. What it establishes is narrower: the bridge of Table 2 is a property of the two standards under $S1'$ and $S2$, not an artifact of the order in which the curator happened to write the crosswalk down.

5. Do existing systems produce this bridge?

Mature systems exist for both halves of this problem: matchers that propose correspondences, and repair facilities that restore coherence. We ran the natural incumbents on exactly this material and evaluated every output with the same HerMiT harness and criteria. Three questions: does a state-of-the-art matcher *find* the expert backbone; is its output *safe* to consume as logic; and, given the expert crosswalk, does an established *repair* facility recover what orientation search recovers?

5.1. Classical matchers on the pair

We ran LogMap and LogMapLt (July-2021 standalone, `logmap-matcher-4.0.jar`, default configuration) on the vendored IES–HQDM pair and LogMap on IES–BFO, and evaluated each output, read as axioms per the system's own asserted relations, against $S1'$ and $S2$. LogMap and LogMapLt each recover 3 of the 10 backbone targets and each carry two mappings onto natively unsatisfiable HQDM classes; the full scores are in Table 4, Appendix A.

Three observations, each of which independently justifies a reasoner in the loop.

First, **both matchers accept the false friend** $\text{ies:Event} \equiv \text{hqdm:event}$ and neither finds $\text{ies:Event} \sim \text{hqdm:activity}$. The consequence is not merely a wrong pair: read as logic, LogMap’s output entails $\text{hqdm:activity} \sqsubseteq \text{hqdm:event}$ *inside HQDM*, so every durative activity becomes an instantaneous boundary, an S2 violation against an ontology the matcher was meant to align and not rewrite. Both also accept vacuously true mappings onto hqdm:asset and $\text{hqdm:agreement_execution}$, two of the 39, and both map RDF(S) vocabulary itself ($\text{rdfs:Class} \equiv \text{hqdm:class}$; LogMapLt adds $\text{rdfs:subClassOf} \equiv \text{hqdm:causes_by_class}$). Whether rdfs:Class is visible as a mappable entity depends on the loading path, since IES types its classes with rdfs:Class rather than owl:Class ; our harness normalises that typing and excludes the RDF(S) meta-vocabulary from both signatures, so these correspondences never enter our S2 counts.

Second, **discovery on this pair is genuinely hard, which is the point of offering it as a test case**: the backbone correspondences are exactly the ones lexical evidence cannot find. On IES–BFO LogMap recovers none of the four, its only substantive mapping being $\text{ies:Entity} \equiv \text{bfo:entity}$, the bare root-label match both LLMs also produce (§5.3), and even two mappings suffice for two conservativity violations. Matching from scratch is harder than certifying an expert crosswalk and we do not claim to out-match matchers; the like-for-like comparison is repair, next.

Third, **the 88 new unsatisfiable classes are not a failure of LogMap’s repair but the price of an input assumption that HQDM violates**. Repair systems in this tradition assume the union of the inputs contains no unsatisfiable class [4], which is a reasonable assumption about published ontologies and a false one about this pair. The two explanations are separable, so we separated them, by re-evaluating LogMap’s own output with its two mappings onto natively unsatisfiable HQDM classes removed (`reasoning/logmap_unsat_attribution.py`): new unsatisfiability falls from 88 to 0. All 88 are downstream of the violated assumption rather than of incomplete repair, and on this pair LogMap’s repair does its job. For LogMapLt, which performs no repair at all, the same two mappings account for only 53 of its 217.

That measurement turns up something else, which cuts against coherence-only evaluation generally. Removing those two mappings *raises* LogMap’s conservativity violations from 82 to 200. None of the 82 disappears, so the entire rise is 118 violations that appear only after the removal, and every one of them mentions a class that was unsatisfiable before it. An unsatisfiable class has no non-trivial subsumptions to report and is excluded from a conservativity count, so incoherence *masks* conservativity violations rather than compounding them: a pipeline that repairs coherence and stops there can watch its conservativity numbers worsen as it improves. Measure the two together, and read neither alone. The exclusion cannot flatter the certified bridge, since a new subsumption under an unsatisfiable superclass makes its subclass unsatisfiable too and S1’ catches that; it does mean the 218 of §4.2 is a lower bound.

5.2. LogMap’s repair facility versus orientation search

We gave LogMap’s DEBUGGER facility the same input orientation search received, the ten expert class correspondences promoted to equivalences over the complete IES and HQDM, with module extraction and its post-repair Hermit check enabled. The outcome is instructive on both sides. LogMap keeps all ten mappings and its output is *certifiably safe*: our harness confirms 0 new unsatisfiable classes and 0 conservativity violations. But it achieves safety by one uniform move, weakening every equivalence to the same direction $\text{hqdm:X} \sqsubseteq \text{ies:Y}$, discarding the entire forward content of the crosswalk. Against the certified bridge of Table 2, three things separate its output from ours. All five provable equivalences are lost, LogMap keeping only the reverse half of each. On the four divergent pairs it keeps the semantically wrong direction: $\text{hqdm:individual} \sqsubseteq \text{ies:Entity}$ asserts that every HQDM individual, activities included, is an IES entity, exactly what IES denies by separating ies:Event from ies:Entity ; the assertion escapes detection only because IES declares no disjointness, and survives S2 only because LogMap simultaneously dropped the forward edges that would have exposed it. And the mapping onto the broken class is kept unflagged, $\text{hqdm:participant} \sqsubseteq \text{ies:EventParticipant}$ subsuming a provably empty class, where orientation search weakens the target one hop to hqdm:state and keeps a contentful axiom. Appendix A gives each with its witness.

The comparison is therefore not repair-fails-versus-repair-succeeds; both outputs are coherent and conservative. It is *retained logical content at equal safety*: 10 uniformly reversed axioms against 21 directed ones with five class and three property equivalences, the expert’s forward direction wherever safe, and a counterexample attached to every refusal.

5.3. An LLM oracle

We also probed two LLM oracles as a ranking task against the certified bridge, 12 queries to Qwen3-Coder-30B-A3B-Instruct at 8-bit and to an IES-specialised fine-tune, scoring 6/12 and 7/12. At that size the aggregate is a probe and not a benchmark, but the error structure is consistent: both pick `hqdm:event` for `ies:Event` and neither reaches `bfo:history`, both answering `bfo:entity`. An LLM oracle is accurate where lexical evidence suffices and fails where the standards diverge, which is where the alignment is hard: a reasonable candidate generator, a poor adjudicator of upper-ontology disagreement. Appendix D has the setup, scores, error structure and why we doubt richer prompting repairs these failures [26].

6. Discussion

Publish the coherence report with the test case. The 39-class defect in `hqdm.owl` was invisible to everything except a DL reasoner run on the input alone; it cost nothing to find and changes how any alignment into HQDM should be read. A matching test case should ship a coherence report on each input, and a reference alignment should flag any correspondence whose target is unsatisfiable: precision and recall over such correspondences measure agreement about the empty set.

Why align into an ontology one could repair instead. The obvious objection is to repair the input first, and §4.1 answers it by measurement: pre-repairing HQDM leaves the 218 conservativity violations exactly where they were and returns the identical bridge, so the policy meets the same problem one step later. The rest is about who may repair what. `hqdm.owl` is the published form of a national-infrastructure data model shipped under Crown Copyright, and a private fork with 39 classes deleted is a different ontology nobody else is using. The defensible position is not that incoherent inputs should be aligned rather than fixed, but that an evaluation should say which it is doing, which is all a coherence report asks. The DISO track [25] moves the campaign into a domain where the inputs are as-shipped government standards, which makes the case likely rather than hypothetical.

Direction is information, and alignment formats should carry it. As §4.3 shows, one-directional subsumption is often the true relation between standards built by different communities. SSSOM and EDOAL can already express directed correspondences; what is missing is the habit of computing the direction rather than defaulting to equivalence, and orientation search is one way to compute it. There is a stronger version of the same idea that we did not take. The crosswalk’s six documented divergences are carried as low-confidence `skos:relatedMatch` warnings, but a divergence a curator is confident about is more honestly stated as disjointness than as a weak match. SSSOM can carry `owl:disjointWith` as the mapping predicate, and declaring the divergences that way would put them under the same reasoner scrutiny as the mappings. It would also give a principled reading to a matcher output that leads to logical error: such a pair is evidence for a disjointness claim, not only a mistake to discard.

Relevance to OAEI. The DISO track introduced in the OAEI 2026 campaign [25] evaluates alignment over defence, intelligence and security ontologies, and ships both a repaired and an unrepaired reference alignment ($R_{\approx}^+$ and $R_{\approx}$), making coherence a first-class concern of the evaluation rather than a post-processing step. This paper’s material has that same two-reference structure, produced by the method itself: the naively promoted crosswalk of §4.2 is an unrepaired reference (consistent, incoherent, 218

conservativity violations) and the certified bridge of §4.3 is its repaired counterpart, the delta between them fully explained by machine-found counterexamples. DISO ships IES v5, BFO and JC3IEDM but not HQDM, so the IES–HQDM leg is the join the collection cannot currently evaluate, and the pair has a profile a track wants: small enough for exhaustive reasoner-based evaluation, adversarial for lexical matchers (§5.1), with an exact-label false friend as its hardest correspondence. We offer the three ontologies, the crosswalk with its six divergences and both reference alignments as a candidate test case, with one property the track does not yet score: the repaired reference is *directed*, each non-equivalence carrying its counterexample, so a system can be scored not only on which pairs it finds but on whether it gets the logical strength of each right. The organisers have signalled an intention to bring HQDM into the collection, and we would rather contribute the material than publish it beside the campaign.

Two threads we have not pursued follow from §5. One is oracle-guided repair-plan selection: where several orientations survive certification the choice is currently made by greedy order, and a human or an LLM oracle in the sense of Lushnei et al. [26] is the obvious adjudicator, §5.2 being precisely a case where the safe answer a system returns is not the one a curator wants. The other is the rest of IES: `ies:Entity` reaches BFO through `bfo:history`, but much of `ies-common` is domain vocabulary about people, organisations and assets whose nearest realist counterparts sit in the Common Core Ontologies above BFO rather than in BFO itself, a larger problem than the 4D backbone treated here.

7. Limitations

The crosswalk is a 17-correspondence backbone, not a full alignment: above it the powertype hierarchies stop being parallel and become a genuine matching problem we have not attempted. The correspondences were hand-curated by a single author, so what is certified is the *logical form* of each, not its correctness as a claim about the domain.

Orientation search is greedy and offers none of the minimality guarantees of the diagnosis-based repair tradition (§3.3). §4.6 measures what that costs here and finds it nothing, but 17 correspondences over two upper ontologies is a small and unusually well-behaved instance: IES contributes no disjointness, so accepted axioms interact sparsely. Nothing shows the operator would be order-insensitive on a large automatically generated alignment where mappings interact densely, and there a procedure that accepts the first correspondence it is handed, however much later weakening that forces, would be the wrong design. Weakening also walks one ancestor chain rather than the full lattice and cannot re-target across branches (§4.4); making re-targeting part of the search, through a candidate generator over the target ontology filtered by the same two criteria, is the obvious next step and we have not taken it.

The systems comparison has its own fairness bounds: the July-2021 LogMap standalone in default configuration; the DEBUGGER fed the ten class correspondences only; AgreementMakerLight [21] and BERTMap [10] not run. Extending to them, and to the conservativity-repair algorithms of Solimando et al. [5, 24], is the natural next experiment. The LLM probe is 12 queries on two models from one family, which supports the qualitative claim about error structure and no quantitative claim at all. Finally, we certify against HermiT alone. Every result was re-checked by invoking the HermiT command-line tool directly on the serialised merges, bypassing owlready2’s object model, and it agrees exactly, including HQDM alone at 39 and IES+HQDM with the certified bridge at the identical 39. That is a check on the harness, not an independent prover; a second reasoner would strengthen the certificates.

8. Related work

Alignment incoherence and its repair are the direct ancestors of this work. Meilicke’s analysis of alignment incoherence and the diagnosis-based repair tradition [2, 3] established that a coherent alignment is the object of interest and that repair is a minimal-hitting-set problem over mappings. LogMap [4] made scalable logic-based repair a practical component of a matching system, and it is the closest prior art for the move this paper makes. LogMap represents an equivalence correspondence as two propositional implications and its repair may discard one while retaining the other, so weakening an

equivalence to a subsumption has been in the repertoire for over a decade, and §5.2 shows it happening: the DEBUGGER returns ten uniformly reversed subsumptions, not ten deletions. Re-orientation as such is therefore not our contribution and we do not claim it. What we add is the search over it. LogMap does not propose an alternative, weaker target among the ancestors of the asserted one; it does not attempt completion back toward equivalence once safety has been established; and it decides over a propositional projection rather than a full DL entailment check, so the direction that survives is a by-product of repair rather than a quantity that was searched for. Making direction the search variable, refereed by a complete reasoner against coherence *and* conservativity, with completion toward equivalence and a counterexample attached to every refusal, is the delta, and §5.2 measures what it is worth on this pair.

The conservativity principle and its violation-repair algorithms [5, 24] supplied the second criterion we use, and the theory of conservative extensions and safe ontology reuse [6] is where that criterion comes from. A further difference in setting is worth stating: repair here happens *after* an expert has committed to the correspondence set, so removal would discard curated knowledge, which is what makes direction the interesting free variable and turns the repair log into documentation of where two government standards disagree.

Axiom weakening [7] is the other parent, and Li and Lambrix [20] developed weakening-and-completing for EL ontologies; our P1+P2 applies that shape at the alignment layer, to the mapping target rather than an axiom of either input.

On the machine-learning side, BERTMap [10] established the pattern of a learned matcher followed by a symbolic repair stage, and recent work applies LLMs as candidate generators or oracles [11, 26]; our §5.3 probe is a small negative datapoint for the oracle role exactly where lexical evidence misleads. MELT [12] is the platform the OAEI 2026 campaign runs on. HQDM derives from West’s data-modelling work [13] in the BORO tradition [14]; BFO is described in [15] and standardised as [16]. A companion census of 528 public ontologies [17] found that 56% declare no constraint axioms at all, IES4 among them.

On IES–BFO specifically, Bailey et al. [19], a team including the original authors of IES, BORO and BFO, compare the two frameworks through parallel modelling of competency-question scenarios and propose graph-level mapping patterns, one relating an IES4 4D individual to a BFO material entity *together with its bfo:history*. Our certified bridge sharpens that: `ies:Entity` subsumes safely under `bfo:history` but not `bfo:material entity` (Table 3), so of their pattern’s two legs exactly one survives as an axiom. They explicitly defer provable alignment to future work and this paper is that step for the 4D backbone. Against our earlier SKOS-level sketch crosswalk from JC3IEDM to IES [18], this is, to our knowledge, the first *certified* alignment involving IES and the first public IES–HQDM alignment. Euzenat and Shvaiko [1] remains the reference text.

9. Conclusion

Two upper ontologies from the same intellectual tradition, aligned by an expert on 17 backbone correspondences, produce a merge that passes a consistency check and leaves a hundred operational classes unsatisfiable; one of them had already rendered 39 of its own unsatisfiable before anyone aligned anything. Both facts are cheap to find with a reasoner, invisible without one, and absent from any similarity score.

The repair is not deletion. Treating the direction of each correspondence as the search variable, with a prover refereeing coherence and conservativity at every step, returns a bridge that keeps every expert mapping, states full equivalence wherever the standards agree, attaches a machine-found counterexample wherever they do not, and comes back unchanged under all 200 correspondence orders we tried. No system we tested recovers it. BFO’s 3D core is crossable only through the occurrent branch, because a 4D entity is a continuant’s history, not the continuant.

Declaration on Generative AI

During the preparation of this work, the author used Anthropic Claude in order to: draft and revise text, assist literature triage, and prepare LaTeX. The locally served Qwen3-Coder-30B models in Section 5.3 are systems under evaluation, not writing tools. All experiments, reported numbers and reasoner runs were designed, executed and verified by the author. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the publication’s content.

A. LogMap’s repair output against the certified bridge

This appendix expands §5.1 and §5.2.

Table 4

Matcher outputs on the vendored pairs, evaluated with Hermit under the criteria applied to the certified bridge. “Backbone” scores correct/wrong/missed against the pair’s gold targets, the ten expert class correspondences for IES–HQDM and the four certified targets for IES–BFO; “unsat targets” are accepted mappings onto one of HQDM’s 39 natively unsatisfiable classes.

System (pair)	mappings	backbone	false friend	unsat targets	new unsat	S2 viol.
LogMap (IES–HQDM)	24	3/1/6	yes	2	88	82
LogMapLt (IES–HQDM)	36	3/1/6	yes	2	217	23
LogMap (IES–BFO)	2	0/1/3	—	0	0	2
certified bridge (IES–HQDM)	13	10/0/0	no	0	0	0

- **All five provable equivalences are lost.** Orientation search proves `ies:Element` $\equiv$ `hqdm:spatio_temporal_extent` and four more; LogMap keeps only the reverse half of each.
- **On the four divergent pairs, LogMap keeps the semantically wrong direction.** `hqdm:individual` $\sqsubseteq$ `ies:Entity` asserts that every HQDM individual, including its activities, is an IES entity, precisely what IES denies by separating `ies:Event` from `ies:Entity`. The assertion is undetectable only because IES declares no disjointness axioms, and it survives S2 solely because LogMap simultaneously dropped the forward edges that would have exposed it. Orientation search, which preserves the expert’s forward commitment as its baseline, rejects exactly this direction with a named counterexample.
- **The mapping onto the broken class is kept, unflagged.** `hqdm:participant` $\sqsubseteq$ `ies:EventParticipant` subsumes a provably empty class; coherence checking cannot distinguish vacuous from safe. Orientation search instead weakens the target one hop to `hqdm:state` and keeps a contentful axiom.

B. Witnesses for the rejected reverse edges

These are the four reverse edges orientation search rejected under S2, with the counterexample the reasoner returned for each. Each violation set is stored sorted, and every witness quoted below is among the first six entries of its sorted set, so the citations are reproducible rather than an artifact of the order in which the reasoner happened to enumerate them:

`ies:Entity` $\sim$ `hqdm:individual` (**104 S2 violations**). The reverse forces `ies:Arrest` $\sqsubseteq$ `ies:Entity`, and likewise `ies:AccountAdminEvent`, `ies:AgreementExecution`, `ies:Assessment` and 100 further IES classes. HQDM asserts `hqdm:activity` $\sqsubseteq$ `hqdm:individual`, while IES holds `ies:Event` and `ies:Entity` apart, so every IES happening is dragged under `ies:Entity`. HQDM’s `hqdm:individual` is strictly broader than IES’s notion of a whole-life persisting thing.

`ies:State` $\sim$ `hqdm:state` (**109 S2 violations**). The reverse forces `ies:Arrest` $\sqsubseteq$ `ies:State` by the same route, since HQDM’s `hqdm:individual` sits under `hqdm:state`. An arrest becomes a state of IES.

`ies:PeriodOfTime` $\sim$ `hqdm:period_of_time` (**1 S2 violation**). The reverse forces `ies:PossibleWorld` $\sqsubseteq$ `ies:PeriodOfTime`, because HQDM asserts `hqdm:possible_world` $\sqsubseteq$ `hqdm:period_of_time` and IES does not.

`ies:ParticularPeriod` $\sim$ `hqdm:period_of_time` (**5 S2 violations**). The reverse forces `ies:PeriodOfTime` $\sqsubseteq$ `ies:ParticularPeriod`, collapsing the distinction between a period and an ISO 8601-anchored particular one, and likewise for `ies:ArbitraryPeriod`, `ies:RecurringPeriod` and `ies:PossibleWorld`.

C. Guarantees and failure modes in full

This appendix expands §3.3.

What is guaranteed. CERTIFY decides $S1'$ and $S2$ on every candidate axiom before it enters B , and on B once more at the end, so the returned bridge satisfies both criteria by construction: the operator cannot return an alignment that violates coherence or conservativity. What it can return is a degenerate bridge. A correspondence whose entire ancestor chain fails is discarded, and one weakened far enough up that chain buys safety with an axiom true of almost everything, as walking from `bfo:material_entity` to `bfo:entity` would (§4.4). Failure is therefore visible in the contentfulness of the output rather than in its safety, and the decision log names the move that produced it. Two things the certificate does not claim. $S1'$ is relative, so a bridge over natively incoherent inputs is certified to add no unsatisfiable class rather than to be coherent. And $S2$ is decided as an entailed subsumption between two named classes of one input, the deductive approximation of conservativity used by Solimando et al. [5, 24] and Cuenca Grau et al. [6], not model conservativity.

The repair we did not run. Deleting HQDM’s 39 unsatisfiable classes (§4.1) is the crudest available repair; a maintainer would more likely revisit the 14 disjointness axioms the OWL rendering introduced. We did not run that variant, and the conservativity result does not depend on it: the 218 violations are subsumption entailments carried by HQDM’s asserted hierarchy, in which no disjointness axiom participates, and deleting axioms cannot create entailments that were not already derivable. Note also that the repaired-target bridge of §4.1 reaches `hqdm:state` by reading the ancestor chain from the unrepaired file: a curator working from the repaired target alone would never have written that correspondence. The published file calls itself “an approximation ... generated with known consistency issues”, and is also the only form a matcher can consume.

Relation to minimal repair. Orientation search optimises neither minimality notion the repair literature uses. LogMap [4] computes a local minimal repair, minimising the mappings removed or weakened per unsatisfiability; ALCOMO [2] and AgreementMakerLight [21] minimise globally over all unsatisfiabilities at once; Solimando et al. [5, 24] give the conservativity analogue. This operator does neither. It fixes the correspondence set and searches the *direction* of each member, accepting the strongest orientation the oracle certifies, so what it conserves is semantic strength per correspondence rather than the number of correspondences touched. On a curated crosswalk, where each correspondence carries expert intent and deletion is the expensive move, that is the trade we want. On a large automatic alignment with confidence scores it is not, and minimal repair remains the better-founded objective.

Failure modes. Three are worth naming. First, weakening walks the *ancestors* of the asserted target, so a correct target on a different branch is unreachable; §4.4 is exactly that case, and it is why the four BFO targets were supplied by hand. Second, a correspondence whose entire ancestor chain fails is

discarded, which is the only deletion move the operator has and is a fallback rather than a strategy. Third, the operator is silent about native defects in its inputs: with $S1'$ in the relative form it neither repairs them nor refuses to run in their presence, and §4.1 measures separately what pre-repairing the target would change.

Behaviour when the inputs are coherent. If neither input has an unsatisfiable class, $S1'$ coincides with absolute coherence and nothing else in the procedure changes. In the fragment studied here the first correspondence is then always accepted at its asserted target, because IES declares no class disjointness and a lone correspondence has nothing to contradict; the discriminating work begins at the second correspondence, once accepted axioms start interacting with one another. The same observation is why the first two rows of Table 3 are a control.

D. The LLM oracle probe in full

This appendix expands §5.3.

LLM-as-oracle is now a standard component of matching systems, used to adjudicate candidate correspondences. We wanted a direct test on this alignment, using our certified bridge as gold, in the shape of the DISO-OAEI ranking task: given a source class and a candidate pool from the target ontology, pick the best target or return NIL.

The benchmark is 12 queries (8 into HQDM, 4 into BFO) built from the certified bridge. We evaluate two systems: Qwen3-Coder-30B-A3B-Instruct at 8-bit, and an IES-specialised fine-tune of the same base model, both served locally at temperature 0. Results in Table 5.

Table 5

Mapping-oracle accuracy against the Hermit-certified bridge. $n = 12$.

System	into HQDM (8)	into BFO (4)	total
Qwen3-Coder-30B-A3B-Instruct (8-bit)	5	1	6/12
+ IES fine-tune	6	1	7/12

The aggregate is uninformative at this sample size and we do not claim a significant difference between the two systems. The *error structure* is the result worth reporting, and it is consistent across both:

- **Both models pick `hqdm:event` for `ies:Event`.** This is the false friend the crosswalk documents as its single most dangerous correspondence: a durative happening with participants mapped onto an instantaneous temporal boundary. The lexical pull of the identical label defeats both models, and the fine-tune does not help, because the trap is not a vocabulary problem.
- **Neither model reaches `bfo:history`.** Asked for the BFO counterpart of `ies:Entity`, both answer `bfo:entity`: a shallow label match onto BFO’s root. The certified answer requires knowing that a 4D whole-life individual is an occurrent, which is a commitment neither model retrieves.
- **Both succeed where the mapping is easy.** `hqdm:thing`, `hqdm:state`, `hqdm:period_of_time`, `hqdm:possible_world` and `hqdm:spatio_temporal_extent` are all recovered by both.

The pattern is that an LLM oracle is accurate exactly where a lexical baseline would also be accurate, and fails exactly on the pairs where the two standards genuinely diverge, which is where the alignment is hard and where the value lies. On this evidence, an LLM is a reasonable candidate generator and a poor adjudicator of upper-ontology disagreement; the reasoner is doing work the model cannot substitute for. We report this as a probe with an explicit sample size, not as a benchmark, and we say what would make it one in §7.

One caveat sharpens the reading rather than softening it. Lushnei et al. [26] show that LLM oracles are markedly sensitive to how the task and its context are presented, and that used selectively, to adjudicate only the correspondences a matcher is least sure of, they reach the top of the OAEI 2025 bio-ml ranking. Our prompt is plainer than theirs: the source class arrives with its label and its `rdfs:comment`, the candidate pool as bare target class names without definitions. The numbers here are therefore a floor. What we doubt is that richer context repairs *these* two failures, since both turn on the correct answer contradicting an identical or near-identical label, and a context window containing the string `hqdm:event` does not obviously help a model refuse it. Running specialised models against the DISO network under the Lushnei et al. protocol would settle the question, and we have not done it.

E. The fused three-ontology artifact

This appendix expands §4.5.

The $\text{IES} \leftrightarrow \text{HQDM}$ and $\text{IES} \leftrightarrow \text{BFO}$ bridges are not merely individually safe. Their union over the complete $\text{IES} \cup \text{HQDM} \cup \text{BFO}$ is also certified at 0 new unsatisfiable classes and 0 conservativity violations, and it admits direct $\text{HQDM} \leftrightarrow \text{BFO}$ anchors on top, found the same reasoner-refereed way: `hqdm:spatio_temporal_extent` $\equiv$ `bfo:occurent`, `hqdm:individual` $\equiv$ `bfo:history`, `hqdm:activity` $\sqsubseteq$ `bfo:process`, `hqdm:state` $\sqsubseteq$ `bfo:occurent`, `bfo:temporal_region` $\sqsubseteq$ `hqdm:period_of_time`. The fused artifact is 32 subsumption axioms plus the `cwb:hasHistory` declaration. `hqdm:participant` is deliberately excluded from the $\text{HQDM} \leftrightarrow \text{BFO}$ anchors: it is one of the 39 native casualties, so any mapping onto it would be vacuously true.

That the two independently certified bridges compose without new violations is not guaranteed by their individual certificates; we checked it rather than assumed it. Composition of certified alignments must, in general, be re-certified.

F. Data and code availability

The crosswalk (SSSOM and SKOS/PROV-O), divergences, SHACL shapes, orientation search, the order-robustness, pre-repair and attribution experiments (`order_robustness.py`, `hqdm_prerepair.py`, `logmap_unsat_attribution.py`), the three certified bridges with full decision/counterexample logs, the LogMap/LogMapLt outputs with their evaluation harness (`reasoning/systems/`, `systems_eval.py`), and the LLM benchmark are released under CC-BY-4.0 at <https://github.com/fabio-rovai/ies-hqdm-crosswalk>; everything reproduces with python `reasoning/orientation_search.py` given any JDK. The input ontologies are public (IES: OGL v3; HQDM: Apache-2.0, Crown Copyright; BFO 2020: CC-BY). This work is independent and not affiliated with or endorsed by Dstl, GCHQ, the IES Working Group or the NDTP.

G. Reasoner, software and hardware versions

Every reasoning result in this paper was produced on one machine: Apple M3 Max, 16 cores, 128 GB RAM, macOS 26.5.2 (build 25F84). No GPU is used for the reasoning experiments; the locally served models of §5.3 run on the same machine’s unified memory. The software stack is Python 3.13.14, `rdflib` 7.6.0, `owlready2` 0.51, and the HermiT build bundled with `owlready2`, Implementation-Version 1.3.8.1099, executed on OpenJDK 26.0.2 (Homebrew). The cross-check reported in §7 invokes that same HermiT jar through its command-line interface, `org.semanticweb.HermiT.cli.CommandLine`, on the serialised merges. LogMap and LogMapLt are the July-2021 standalone distribution (`logmap-matcher-4.0.jar`) in default configuration, run on the same JDK with `--add-opens` flags for module access.

The input ontologies are pinned by content hash: `ies-common.ttl` (IES 5.0.3, version IRI <http://informationexchangestandard.org/ont/ies/common/5.0.3/>) `sha256:f742. . . 4b37e5`; `hqdm.owl` from `gchq/HQDM` `sha256:be27. . . 47caff8`; `bfo.owl`, the BFO 2020 class release

from purl.obolibrary.org/obo/bfo.owl, sha256:ca39...6c4666. Full hashes are in reasoning/RESULTS.md in the released repository. The complete orientation search, both bridges and the BFO ablation included, runs in 33 s wall-clock on this machine; the 200-order study of §4.6 takes 19 minutes, most of it saved by memoising the oracle on the trial axiom set.

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